Why HackerFab

Hi HackerFab IITB, I am Dhruv Rao, a first year studying EP (Engineering Physics) in IITB. I knew about HackerFab even before reaching iitb. I watched all videos documenting your progress and have even read the articles published on various websites and newspapers about you. Since then getting into HackerFab has been at the back of my mind as the ideology appealed to me. I genuinely feel like this is the only 'club' which has an actual goal doing something groundbreaking. HackerFab is the only club I have applied to.

From what I understand about HackerFab, it's more a startup than a club. Making a DC sputter for only 7 lakhs is groundbreaking, making a tube furnace for only 30k is groundbreaking, creating a lithography machine with micron level precision is a huge achievement on its own. But the way HackerFab treats all this as just a step towards its ultimate goal of making fabrication accessible really inspires me.

Short Term

Now, coming to what I would do at HackerFab. As a first year in EP I get to attend multiple seminars conducted by various professors and even seniors on vast number of topics. I have always been interested in quantum mechanics and have been researching into its greatest application as of recent times. Quantum computing. Quantum computing is a field which seems shrouded in mystery and it is since research is still being done on the same by techgiants, IBM and Google. But it isn't impossible to get to the cutting edge of quantum computing and I wish to take HackerFab there. Making a Josephson junction, and hence making physical and then logical qubits is a far reach for HackerFab right now. One of the problems we face is that the HackerFab lithography machine's resolution is theoretically 1 micron but I would need to get it down one order of magnitude to make Josephson junctions a reality. So I would like to work on and improve the lithography machine . The docs show about 4 µm/6 µm lines for a nominal 5 µm pattern, while the optics (405 nm, NA 0.25) allow roughly 1 µm. This suggests there is a problem with the process used during lithography which is the cheapest thing to tackle first. Improving k_1 through process control is the cheapest way to improve resolution, before changing any hardware. Then attaining sub micron resolution would require switching out the .25 numerical aperture for a larger .45-.6 NA.

Long Term

Long term, my plan is to help demystify quantum computing at HackerFab IITB using the resources available. Breakthroughs are happening all the time with research on better orientations of qubits to reduce error coming out . Being at the cutting edge of technology leads to way more money than just being a fabrications lab and in the orientation we were told to think in crores. A quantum capable fab would attract research collaborations and grant which will help in the long run. My long-term idea is to fabricate superconducting quantum chips that can be sold to research labs that already have the cryogenic infrastructure to run them.

Every year of my journey will be towards the goal of the team. I would like to learn under the seniors at hackerfab and ,by being more hands on, improve my understanding of all concepts. Nowhere else on campus will I be exposed to the environment here at HackerFab and it's an experience I do not wish to pass on.